How do differences in e-bike legislation across major US cities (Chicago, Los Angeles, New York City, and Atlanta), relate to injury rates, and to what extent might other factors influence these outcomes?

As electric bicycles (e-bikes) become an increasingly popular form of urban transportation, cities face the challenge of keeping up with rider safety, accessibility, and the sustainability benefits of eco-friendly options of transit. E-bikes offer an alternative to automobiles, and more importantly relieve transportation related emissions. They also provide those not physically able enough for pedal bicycles with a viable alternative. However, the rapid rise in e-bikes popularity has outpaced the appropriate legislature needed to regulate them as any other road vehicle requires. Concerns regarding rider safety, speed, and interaction with other drivers on the road have led to a range of different regulations adopted by the different states and cities regarding the use of e-bikes. Our research aims to explore the legislation regarding e-bikes in major cities. We will be focusing specifically on Chicago, Los Angeles, New York, and Atlanta to find out if different regulations are connected with different reported injuries among bicycles and e-bicycles riders. With this information we will examine if different regulations correlate to different reported injuries, while also attempting to account for confounding variables such as city connectivity or bike lane infrastructure.

Regarding sustainability, we'd like to address the social and environmental dimensions for this topic. Social sustainability involves ensuring that e-bikes remain a safe and accessible transportation option for riders of all types. As mentioned earlier, one of the most attractive aspects of e-bikes is their universal accessibility. The elderly, children, and people who are physically unfit would normally have a hard time using pedal bicycles on the road. E-bikes provide a previously inaccessible choice for mode of transport, at a fraction of the cost of a car. The dimension of environmental sustainability is represented by the capacity of e-bikes to contribute towards cutting back on car usage, traffic, and pollution. The economic aspect of sustainability can also come into play with regards to e-bikes being less costly than cars. Not only are e-bikes much less costly than cars for the individual to buy and maintain, they require significantly less resources from governments to accommodate them and their required infrastructure. Car infrastructure costs major cities billions each year to maintain, along with the immense land use that is dedicated to roads and parking infrastructure, which usually makes the city a fraction of what the land could be worth if zoned as commercial or residential. We will

examine how all of these dimensions are reflected in the different legislature of the four selected cities.

Preliminary source(s) we plan to consult:
Chicago: legislation, injuries/crashes, confounding
Atlanta: legislation, injuries/crashes, confounding
Los Angeles: legislation, injuries/crashes, confounding, other
NYC: legislation, injuries/crashes, confounding

Other→ types of bike lanes case study